

A sample Results section: Improving instructor ethos through document design

The example below is from a small research study my colleagues and I did to analyze how document design — the use of visual elements to organize information on a page — affects student perceptions.¹ Our purpose was to learn if the additional time it takes instructors to visually format the classroom materials is really worth the investment. We wanted to know if this design really mattered.

To answer this simple research question, we showed over 160 incoming, first-year university students the original and redesigned versions of the assignment in [Figure 8.5](#). Approximately half of the students saw the original version first and half saw the redesigned version first. All students answered survey questions asking about their perceptions of the instructor and their preferences for one version over the other.

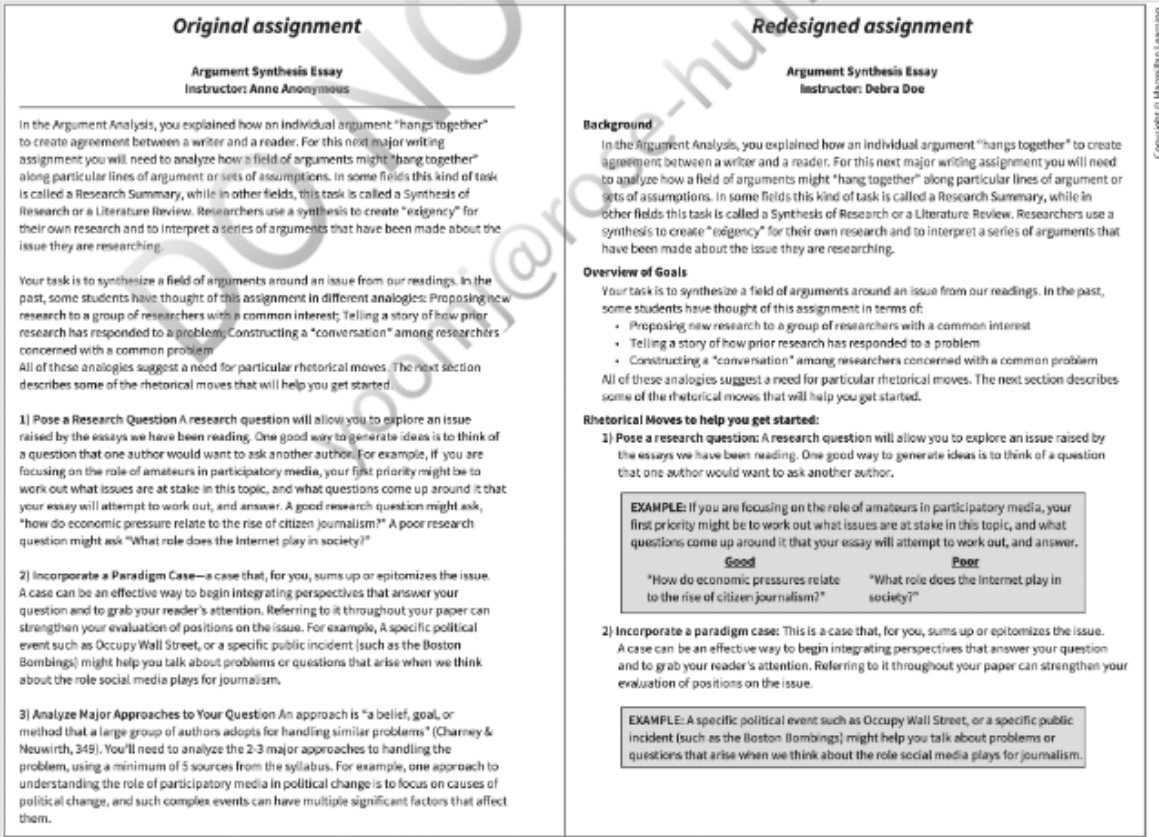


Figure 8.5. Two Versions of the Same Assignment. The redesigned version uses more visual formatting. The purpose of the study was to see if the additional effort of producing the redesigned version was worth the time

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Take a look at an abbreviated version of our Results section and note the types of information we included, how we organized our results, and how we mixed quantitative and qualitative data.

Excerpt from "Improving Instructor Ethos through Document Design"
by Wolfe, Roderick, and Rooney

RESULTS

Students strongly preferred the redesigned prompt

[Figure A.1](#) demonstrates that incoming freshman preferred the redesigned prompt. Overall, 65 percent of our participants indicated that they somewhat or completely preferred the redesigned prompt, whereas only 18.7 percent of participants somewhat or completely preferred the original, a difference that is highly statistically significant, $\chi^2(4) = 34.07, p < .0001$.

Students' preference for the redesigned prompt was consistent regardless of their gender, major, or self-proclaimed interest in writing. There was a slight effect for the order in which students saw the prompts: those who saw the original prompt first were even more strongly in favor of the redesigned prompt than those who saw the redesigned prompt first.

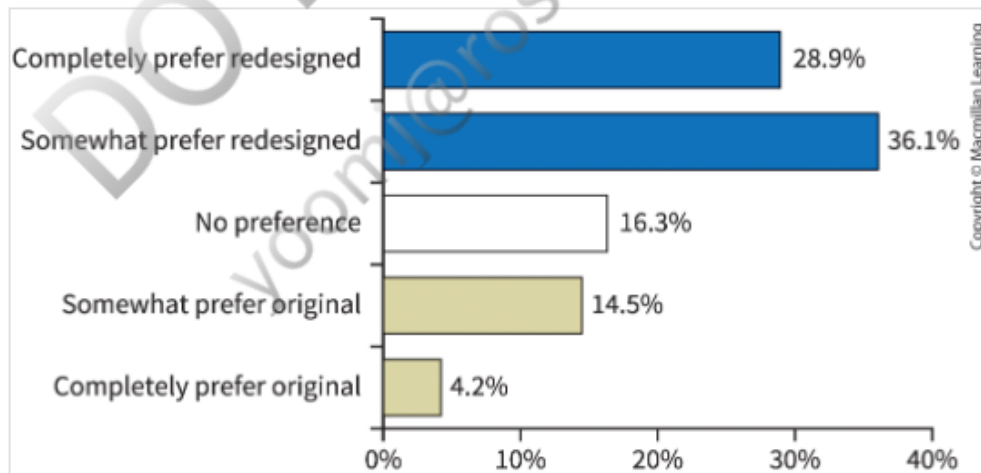


Figure A.1. Student Preference for the Two Assignment Prompts.

In open-ended comments, students noted that the formatting of the redesigned prompt helped them better comprehend the task. They claimed that the distinctions among different types of information helped them "mentally compartmentalize" (Participant 31) different

aspects of the project, gave them a better idea of the task and where to start, and made for generally "less boring" (*Participants 48, 103*) reading than the original prompt. These findings all suggest that the formatting is responsible for students' preference of the redesigned prompt.

Students made inferences about instructor personality from the design

While [Figure A.1](#) shows that students preferred the redesigned prompt and found it easier to read, [Table A.1](#) shows that these preferences translated into a more favorable impression of the instructor who created the prompt. Students were more likely to indicate that the instructor behind the redesigned prompt was knowledgeable, caring, enthusiastic, and experienced.

Table A.1. Student Impressions of Instructors Who Wrote the Prompts on a 4-Point Likert Scale (1 = Not at All; 4 = Very Much).

Survey questions	Original	Redesigned
How knowledgeable do you think this instructor is about academic writing?	3.58	3.70*
How experienced do you think this instructor is?	3.45	3.68**
How much do you think this instructor cares about her students?	3.16	3.39**
How enthusiastic do you think this instructor is about the course?	3.13	3.33**
How comfortable would you feel asking this instructor questions about this assignment?	3.17	3.27

* $p < .05$, ** $p < .01$.

Several students elaborated on these perceptions in their comments, noting the additional effort and care the redesigned prompt required.

[The instructor of the redesigned prompt] seems more likely to be able to answer questions and definitely more willing to, as she spent the time to try and clarify the assignment.

— Participant 13

Her instructions are clearly and consistently formatted. This makes it seem like she really cares about her students: she seems to have put time and thought into the preparation of this assignment so that her students understand it and learn from it. It lends credibility to the instructor and makes her seem knowledgeable.

— Participant 119

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— Participant 119

It is more engaging and gives me a sense that [this instructor]
is more willing to work for her students (Which I'm sure both
would do but just given that sheet).

— Participant 55

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